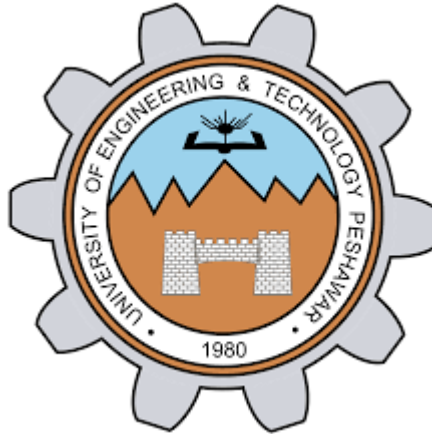




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Assignment 2	Computer programing
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C++ Assignment – Complete Programs (Fully Commented)

```
// ===== QUESTION 1 =====
#include <iostream>
using namespace std;
int main() {
    int arr[10]; // array for 10 integers
    int sum = 0;
    // input values
    for(int i=0; i<10; i++) {
        cout << "Enter value: ";
        cin >> arr[i];
        sum += arr[i]; // add to sum
    }
    int max = arr[0], min = arr[0];
    // find max and min
    for(int i=1; i<10; i++) {
        if(arr[i] > max) max = arr[i];
        if(arr[i] < min) min = arr[i];
    }
    cout << "Maximum: " << max << endl;
    cout << "Minimum: " << min << endl;
    cout << "Average: " << sum/10.0 << endl;
    return 0;
}
// ===== QUESTION 2 =====
#include <iostream>
using namespace std;
// function to reverse array
```

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void reverseArray(int arr[], int size) {
    for(int i=0; i<size/2; i++) {
        int temp = arr[i];
        arr[i] = arr[size-i-1];
        arr[size-i-1] = temp;
    }
}

int main() {
    int arr[5];
    // input
    for(int i=0; i<5; i++) {
        cin >> arr[i];
    }
    cout << "Before reverse: ";
    for(int i=0; i<5; i++) cout << arr[i] << " ";
    reverseArray(arr, 5);
    cout << "\nAfter reverse: ";
    for(int i=0; i<5; i++) cout << arr[i] << " ";
    return 0;
}

// ===== QUESTION 3 =====
#include <iostream>
using namespace std;
int main() {
    int A[3][3], B[3][3], C[3][3];
    // input matrices for(int i=0; i<3; i++)
    for(int j=0; j<3; j++)
        cin >> A[i][j];
    for(int i=0; i<3; i++)
        for(int j=0; j<3; j++)
            cin >> B[i][j];
    // addition
    for(int i=0; i<3; i++)
        for(int j=0; j<3; j++)
            C[i][j] = A[i][j] + B[i][j];
    // output
    cout << "Result Matrix:\n";
    for(int i=0; i<3; i++) {
        for(int j=0; j<3; j++)
            cout << C[i][j] << " ";
        cout << endl;
    }
    return 0;
}

// ===== QUESTION 4 =====
#include <iostream>
#include <algorithm>
using namespace std;
int main() {
    string names[5];
    // input names
    for(int i=0; i<5; i++)
        cin >> names[i];
    sort(names, names+5); // sort alphabetically
    cout << "Sorted Names:\n";
    for(int i=0; i<5; i++)
        cout << names[i] << endl;
    return 0;
}

// ===== QUESTION 5 =====
#include <iostream>
using namespace std;
float totalLoad(float arr[]) {
    float sum = 0;
    for(int i=0; i<24; i++) sum += arr[i];
    return sum;
}

float peakLoad(float arr[]) {
    float max = arr[0];
    for(int i=1; i<24; i++)
        if(arr[i] > max) max = arr[i];
    return max;
}

int main() {
    float load[24];
    for(int i=0; i<24; i++)
        cin >> load[i];
}

```

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cout << "Total Load: " << totalLoad(load) << endl;
cout << "Peak Load: " << peakLoad(load) << endl;
return 0;
}
// ===== QUESTION 6 =====
#include <iostream>
using namespace std; void findTopper(string names[], int marks[], int size) {
    int maxIndex = 0;
    for(int i=1; i<size; i++)
        if(marks[i] > marks[maxIndex])
            maxIndex = i;
    cout << "Topper: " << names[maxIndex] << endl;
}
int main() {
    string names[5];
    int marks[5];
    for(int i=0; i<5; i++) {
        cin >> names[i];
        cin >> marks[i];
    }
    findTopper(names, marks, 5);
    return 0;
}
// ===== QUESTION 7 =====
#include <iostream>
using namespace std;
float totalEnergy(float arr[]) {
    float sum = 0;
    for(int i=0; i<24; i++) sum += arr[i];
    return sum;
}
int main() {
    float energy[24];
    for(int i=0; i<24; i++)
        cin >> energy[i];
    cout << "Total Energy: " << totalEnergy(energy) << endl;
    return 0;
}
// ===== QUESTION 8 =====
#include <iostream>
using namespace std;
int main() {
    float voltage[20];
    int unsafe = 0;
    for(int i=0; i<20; i++) {
        cin >> voltage[i];
        if(voltage[i] < 3.0 || voltage[i] > 4.2)
            unsafe++;
    }
    if(unsafe > 0)
        cout << "Warning: Unsafe readings exist\n";
    else
        cout << "All readings are safe\n";
    return 0;
}
// ===== QUESTION 9 =====
#include <iostream>
using namespace std;
int main() {
    float grid[4][7];
    float total = 0;
    for(int i=0; i<4; i++)
        for(int j=0; j<7; j++) {
            cin >> grid[i][j]; total += grid[i][j];
        }
    cout << "Total Consumption: " << total << endl;
    return 0;
}
// ===== QUESTION 10 =====
#include <iostream>
using namespace std;
int main() {
    int attendance[5][7];
    int total = 0;
    for(int i=0; i<5; i++)
        for(int j=0; j<7; j++) {
            cin >> attendance[i][j];

```

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total += attendance[i][j];  
}  
cout << "Total Attendance: " << total << endl;  
return 0;  
}
```